Inputs accepted

Create private unique staging directory

Write payload, manifest, and checksums

Validation includes:
base identity
namespace collisions
payload bounds
annotation consistency
complete file set

Identity and completeness valid?

No

Quarantine failed staging artifact

Yes

fsync and atomic rename

Immutable complete artifact

Checksum-pinned deployment selection

Executable mapping canary passes?

Yes

No

Select validated fallback

Activate candidate